

3(a)(i)	semi-lunar valve ; accept aortic valve / pulmonary valve	1
3(a)(ii)	1 from the atria to the ventricles ; 2 into the (left and right) atria from the pulmonary vein and vena cava ; A venae cavae for vena cava A superior vena cava and inferior vena cava	2
3(b)	<i>stage of cardiac cycle:</i> ventricular systole ; <i>reason:</i> bicuspid / mitral, valve <u>and</u> tricuspid valve closed or atrioventricular valves closed or semi-lunar valves / aortic <u>and</u> pulmonary valves, open ;	2

Question	Answer	Marks
3(c)	any three from: 1 impulses reach the atrioventricular node ; 2 short delay / after 0.1 s ; 3 (AVN) sends impulses through, Purkyne fibres / Bundles of His ; A Purkinje for Purkyne 4 impulses (travel down through the septum and) reach the, base / apex, of the ventricles / heart ; 5 atria complete contraction / atria and ventricles do not contract at the same time / ventricles fill with blood / atria empty ; 6 impulses travel upwards / AW, causing ventricles to contract from base upwards ; 7 ref. to (both) ventricles contract at same time ;	3

Question	Answer	Marks
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